
MTH 4330 Group 3 Final Report (NYC Payroll)

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Abstract

This project analyzes employee compensation patterns in the NYC Citywide Payroll dataset, with a focus on predicting total earnings and understanding what best predicts compensation across agencies. We define the target variable, *Total Paid*, as Regular Gross Paid plus Total OT Paid. The project's main goal is a supervised regression task: given employee-level payroll features such as base salary, regular hours, overtime hours, and fiscal year, the model predicts total compensation. We compare a linear regression baseline with a random forest regressor using the same held-out evaluation design. The final results show that the random forest model substantially outperforms the linear baseline, reducing MAE from 4879.505 to 1572.433 and RMSE from 7125.186 to 3313.735, while increasing R^2 from 0.8767656 to 0.973345. These results suggest that nonlinear relationships, especially those involving overtime and salary variables, are important in payroll prediction. Error analysis of our linear baseline shows that agencies with above-average total compensation, such as DISTRICT ATTORNEY-SPECIAL NARC, remain difficult to predict. This may be because it tends to underpredict the pay of employees whose actual pay is substantially higher than average. Overall, the project shows that machine learning can model payroll data effectively, but results must be interpreted carefully because the target variable is closely related to some input features.

1. Introduction

Our project analyzes employee compensation patterns in the NYC Citywide Payroll dataset, with a particular focus on understanding total earnings across agencies.

The main objective of this final project is a supervised regression task. Given employee-level payroll features, we predict *Total Paid*, where:

$$TotalPaid = RegularGrossPaid + TotalOTPaid.$$

This target represents total direct compensation from regular

earnings and overtime. This definition allows us to study how factors such as base salary, regular hours, overtime hours, agency, title, and work location explain differences in total compensation.

The project has two connected goals. The main goal is prediction: we evaluate how accurately different models predict *Total Paid*. The secondary goal is interpretation: after prediction, we aggregate observed and predicted values at the department and individual levels to identify patterns in agencies whose compensation patterns are not fully explained by the model.

We continue segmenting the dataset by pay basis because employees paid hourly, daily, annually, and prorated annually have different compensation structures. In this final report, the numerical results are based on the Prorated Annual subset, which has a manageable size and enough records to compare the baseline and machine learning models.

Our final results show that the random forest model performs much better than the linear regression baseline. The random forest reduces MAE from 4879.505 to 1572.433 and improves R^2 from 0.8767656 to 0.973345. This suggests that payroll compensation is not explained fully by a simple linear relationship.

GitHub repository link for codes and datasets:

<https://github.com/rcheung2/2026-Spring-MTH-4330-Project.git>

2. Background

Public payroll reporting shows that in several major cities, including New York City, many of the highest-paid employees also work large amounts of overtime. This supports our decision to combine regular gross pay and overtime pay into one response variable. Instead of treating overtime as a side measure, our project treats it as a central part of compensation differences (Matthew E. Kahn & Ravi, 2019).

The expansion of Checkbook NYC as a transparency tool is also relevant to our project (New York City Comptroller, 2017). Payroll data is large and difficult to interpret directly, so transparency tools and analytical summaries help make public spending easier to understand. Our project follows a similar goal by using machine learning to summarize and

predict compensation patterns.

Policy discussions around public reporting also emphasize the importance of meaningful spending measures rather than raw data alone (Citizens Budget Commission, 2024). This connects to our project because our goal is not only to report payroll values, but also to explain which features help predict compensation and where the model struggles.

From a machine learning perspective, this project is a tabular regression problem. Linear regression is useful as a baseline because it is simple and interpretable, but it assumes a mostly linear relationship between the predictors and the target. Random forests are better suited for nonlinear tabular data because they can capture interactions between features such as base salary, overtime hours, department, and title. Compared with the intermediate report, the final version places more attention on fair evaluation, clearer model comparison, and the limitations of predicting a target that is closely connected to salary and overtime inputs.

3. Dataset

Our dataset is the **NYC Citywide Payroll** dataset from NYC Open Data. The payroll data comes from the City’s Personnel Management System and is entered by the relevant agencies. Each row corresponds to an employee payroll record for a fiscal year. As noted on the dataset page, the data can be used to study how city financial resources are allocated and how much of the budget is devoted to overtime (Office of Payroll Administration (OPA), 2025).

3.1. Final Dataset Setup

The final experiments focus on the Prorated Annual subset. It was chosen because it was small enough to be uploaded to GitHub, which guaranteed that everyone in our group was using the exact same data. Based on the characteristics of prorated pay, it should be the most responsive to changes in hours worked, which is central to our project. It still contains well over 20,000 data points, which is sufficient for model testing, and has the additional benefit of loading much faster than the other pay-basis categories, which sped up our turnaround time.

Table 1. Final dataset setup for the main experiments.

Component	Description
Dataset	NYC Citywide Payroll
Subset	Prorated Annual
Rows	29,466
Target	Total Paid
Main task	Regression

After cleaning, the training set has 20,626 rows, the devel-

opment set has 4,419 rows, and the test set has 4,421 rows, for a total of 29,466 rows.

3.2. Variables

Our models focus on these variables:

- **Regular Gross Paid:** employee’s total base pay for the fiscal year (Part of Total Paid)
- **Total OT Paid:** employee’s total overtime pay for the fiscal year (Part of Total Paid)
- **Fiscal Year:** Fiscal year. Ranges from 2014 to 2025 in the data we used (Input variable)
- **Base Salary:** employee’s assigned base salary (Input variable)
- **Regular Hours:** regular hours worked during the fiscal year (Input variable)
- **OT Hours:** overtime hours worked during the fiscal year (Input variable)

The target variable is:

$$Y = RegularGrossPaid + TotalOTPaid.$$

We used categorical variables for grouping and post-training analysis:

- **Agency Name:** department or agency name (used for grouping during error analysis)
- **Pay Basis:** prorated annual, per hour, per day, or per annum. Used to split the data into subsets.

3.3. Changes Since the Intermediate Report

The intermediate report already defined the target variable, cleaned core payroll variables, built a linear regression model, built a random forest model, and started department-level residual analysis. The main changes in the final version are related to evaluation clarity and model comparison.

First, the baseline and random forest are described as being compared under the same held-out evaluation structure. This is important because the baseline should not be trained on the full dataset while the random forest is evaluated separately. Second, the development set is now explicitly used for model and hyperparameter selection. Third, the final report separates completed experiments from future work, so the report reads as a finished project rather than an in-progress update.

3.4. Data Cleaning and Encoding

Rows with missing core pay values are removed, and missing overtime values are treated as zero when appropriate.

We avoided directly using categorical variables in the models. Their role was to categorize our data points by department for error analysis and split the dataset into coherent subsets to avoid mixing incompatible pay bases.

4. Methods

We implemented the pipeline around one baseline model and one machine learning model. The baseline model is linear regression, and the main machine learning model is a random forest regressor.

4.1. Evaluation Design

To make the model comparison fair, both models use the same split:

- 70% training set
- 15% development set
- 15% test set

The training set is used to fit the models. The development set is used for feature choices and hyperparameter tuning. The test set is used for final evaluation. All reported performance values in the final results table are treated as held-out test metrics.

Both models are evaluated using mean absolute error (MAE), root mean squared error (RMSE), and ordinary R^2 . MAE gives us an average dollar error, RMSE penalizes large prediction errors more strongly, and ordinary R^2 measures how much variation in total paid is explained by the model on held-out data. Even though R^2_{adj} is standard for linear regression, using ordinary R^2 for both will give us more directly comparable results.

4.2. Feature Engineering

The main response variable is:

$$TotalPaid = RegularGrossPaid + TotalOTPaid.$$

The linear regression and random forest models both use these input variables to predict *TotalPaid*:

- **Fiscal Year**: we encode 2014 as 0 to center it around the earliest year in the dataset. 2015 becomes 1, 2016 becomes 2, etc.
- **Regular Hours**: regular work hours

- **Base Salary**: assigned base salary
- **OT Hours**: overtime hours

Pay basis remains important because income is calculated differently for different pay groups, so we use this to separate the data into multiple tables. For this final report, the main experiment focuses exclusively on the Prorated Annual group.

Agency Name appears to be standardized because it reflects the name of the user agency that submitted the data, so it is sufficient to use it as-is for categorization and error analysis after training the models.

4.3. Linear Regression Baseline

The baseline is a linear regression model that predicts *Total Paid* from our selected numerical input variables. This model is useful because it is simple and interpretable. It shows the relationship between each predictor and the target under a linear assumption.

The model is trained on the training set and evaluated on the held-out test set. This fixes the main evaluation issue from the intermediate report, where the baseline was not directly comparable to the random forest.

The input variables are:

- **Fiscal Year**: centered around the earliest year in the dataset, which is 2014
- **Regular Hours**: regular work hours
- **Base Salary**: assigned base salary
- **OT Hours**: overtime hours

The linear model coefficients are shown in Table 2.

Table 2. Linear regression coefficient table.

Variable	Estimate	Std. Error	t value
(Intercept)	-14907.2226	139.5270	-106.84
Fiscal_Year	-359.4421	15.7859	-22.77
Regular_Hours	19.1545	0.0815	235.06
Base_Salary	0.6054	0.0034	176.67
OT_Hours	9.2955	1.1195	8.30

As expected, Regular Hours, Base Salary, and OT Hours are positively correlated with *Total Paid*. The negative coefficient for Fiscal Year may indicate that this variable is not behaving like a simple inflation proxy in our data.

We can look at the employee-level scatter plot comparing the linear model's predictions to the actual values in Figure

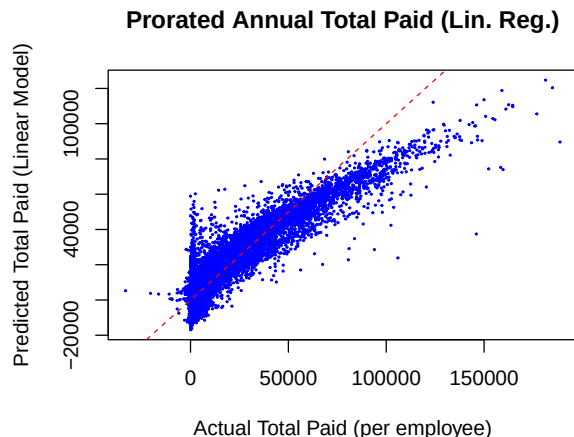


Figure 1. Employee-level actual and predicted Total Paid, evaluated over the entire pre-split Prorated Annual subset.

1. It shows most points follow a pattern that is analogous to a square root or logarithmic graph. This shows that there are some nonlinear relationships that our linear baseline is unlikely to fully capture.

4.4. Random Forest Model

The random forest model is used as our main machine learning method. We chose this model because it can capture nonlinear interactions between pay variables, hours, and department categories without requiring a rigid linear form. This is useful for payroll data because overtime-heavy departments may behave differently from departments with mostly fixed salaries.

At a high level, the pipeline is:

1. input one employee payroll record,
2. extract numerical payroll features,
3. apply the regression model,
4. output a predicted total compensation value, and
5. aggregate results by department for interpretation.

4.5. Hyperparameter Choices

The random forest hyperparameters are evaluated using R^2 on the development set. We compare settings for the number of trees, maximum tree depth, and minimum leaf size in Table 3.

At the extremes, these hyperparameter selections range from underfitting to performing similarly to a balanced set of parameters, but with more overfitting risk.

Table 3. Random forest development-set hyperparameter tuning.

Trees	Max Depth	Min Leaf	R^2
100	8	10	0.967529
200	12	5	0.9713
300	15	2	0.97133

Table 4. Final test performance on the Prorated Annual subset.

Method	MAE	RMSE	R^2
Linear baseline	4879.505	7125.186	0.8767656
Random forest regressor	1572.433	3313.735	0.973345

Our final random forest uses:

- `n_estimators` = 200
- `max_depth` = 12
- `min_samples_leaf` = 5

This selected model balances predictive accuracy with overfitting control.

5. Experiments

The first experiment compares the linear regression baseline against the random forest model. The second experiment uses department-level aggregation to study where the model performs well or poorly.

5.1. Final Performance Results

Table 4 compares the MAE, RMSE, and R^2 of the linear regression baseline and the random forest regressor on the test set.

The random forest model substantially outperforms the linear baseline. MAE decreases by about 3307 dollars, and RMSE decreases by about 3811 dollars. This is strong evidence that nonlinear relationships improve prediction.

5.2. Why the Random Forest Performs Better

The random forest performs better because payroll patterns are not perfectly linear. Overtime can create irregular jumps in total compensation, and departments may differ in how overtime and base salary combine. A linear model can only estimate one fixed relationship for each feature, while the random forest can split the data into smaller regions and model different relationships for different types of employees.

Feature importance also supports this interpretation. The predictors from our baseline model, ranked by absolute t-value, are:

- Regular Hours: 235.06
- Base Salary: 176.67
- Fiscal Year: 22.77
- OT Hours: 8.3

Judging by the t-values, each factor has predictive value, although the influence of overtime hours is relatively small. These features make sense because total compensation is directly connected to salary, regular work, and overtime. However, this also creates an important limitation: because *Total Paid* is constructed from pay-related variables, some of the model's success may come from learning a near-mechanical payroll relationship rather than a deeper causal pattern.

5.3. Department-Level Results

To interpret model performance beyond individual predictions, we aggregate observed and predicted values by department in Figure 2. This helps identify agencies whose compensation patterns are not fully explained by the model.

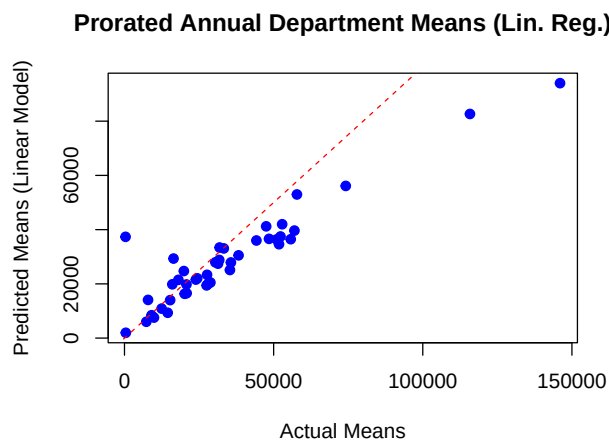


Figure 2. Department-level actual and predicted means, evaluated over the entire pre-split Prorated Annual subset.

While many departments are predicted reasonably well, some agencies deviate from the general pattern. Table 5 shows departments with the largest residuals.

The largest residual belongs to DISTRICT ATTORNEY-SPECIAL NARC. Its actual mean is 146019.73, while its

Table 5. Top departments, sorted by residual. Aggregated from the entire Prorated Annual dataset. (pre-split)

Agency Name	Actual Mean	Residual
DISTRICT ATTORNEY-SPECIAL NARC	146019.73	52004.824
DEPT OF HEALTH/MENTAL HYGIENE	357.22	-36986.684
Spec Narc-DA	115819.94	33163.601
DEPARTMENT OF BUSINESS SERV.	55734.57	19263.345
DISTRICT ATTORNEY KINGS COUNTY	74179.83	18062.044
DISTRICT ATTORNEY QNS COUNTY	56970.17	17325.579
DEPT OF CITYWIDE ADMIN SVCS	51801.4	17182.984
DEPT OF PARKS & RECREATION	52302.51	14828.277
OFFICE OF THE COMPTROLLER	50992.71	14608.313
BRONX DISTRICT ATTORNEY	16439.47	-12891.142

predicted mean is 94014.90. This suggests that the agency has compensation patterns that are not fully captured by the basic model features.

DEPT OF HEALTH/MENTAL HYGIENE appears to have a small mean, but it is also possible that most of its employees are paid on a basis other than prorated annual, so it is more likely that this department is underrepresented in our sample.

6. Conclusion

6.1. Error Analysis

The random forest model improves prediction compared with linear regression, but some departments remain harder to predict. The largest errors should be most likely to appear in departments with unusual compensation structures, specialized roles, or high overtime rates.

At the individual level, likely sources of error include:

- unusually high overtime hours,
- high-salary outliers,
- part-year or seasonal workers,
- job titles with different pay behavior than the department average, and
- agencies with atypical pay rules.

DISTRICT ATTORNEY-SPECIAL NARC case is one example of a department that the model underpredicts by a large amount. It is possible that either department-level averages can hide important title-level or role-level differences, or that the linear model simply has difficulty predicting high earners.

6.2. Strengths and Limitations

The main strength of the project is that it provides a clear prediction framework for a large public payroll dataset. The random forest model performs strongly and captures nonlinear effects that the linear baseline misses.

The main limitation is that the target variable is closely related to the input features. Since *Total Paid* is constructed from regular gross pay and overtime pay, and the model uses variables such as base salary and overtime hours, the prediction problem has a partly mechanical structure. This means that high performance should not be interpreted as proof that the model explains every deeper cause of compensation differences.

Another limitation is that the main experiment focuses on the Prorated Annual subset. Even though it is sufficient for the scope of our project, we are not able to say anything conclusive about underrepresented departments such as DEPT OF HEALTH/MENTAL HYGIENE.

6.3. Lessons Learned and Future Work

This project shows that fair evaluation design is just as important as model choice. The final version improves on the intermediate report by using the same held-out evaluation structure for both models and by making the role of the development set explicit.

The project also shows that random forests are useful for tabular payroll data because they can capture nonlinear relationships. However, the results must be interpreted carefully because a model can perform well by learning direct payroll relationships rather than broader compensation structure.

Future work could include:

- testing models without OT Hours to see how much performance depends on overtime information,
- applying $\log(\text{TotalPaid} + 1)$ to reduce the influence of extreme earners,
- grouping employees by job title or broader role category,
- comparing results across all pay-basis groups, and
- adding more detailed error analysis for high-overtime departments.

Overall, the project demonstrates that machine learning can predict employee compensation effectively while also helping identify agencies and employee groups that deserve closer analysis.

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